

A Hybrid BERT and Metadata Deep Learning Framework for Robust Twitter Bot Detection

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Abstract—Twitter has emerged as a leading platform for instant communication, information sharing, and public interaction, but its open nature also makes it highly susceptible to misuse by automated accounts, or bots. These accounts are often employed to manipulate conversations, spread misinformation, and influence public perception, which compromises the reliability of online discourse. To address this challenge, this paper introduces a Hybrid BERT with Metadata deep learning framework that integrates semantic features from tweet content with behavioral attributes such as follower count, posting activity, and verification status. By combining linguistic and structural signals, the model achieves more reliable and comprehensive classification than approaches that rely exclusively on text or metadata. The framework is trained and tested on the TwiBot-20 benchmark dataset, where focal loss is applied to mitigate class imbalance and AdamW optimization ensures stable convergence. Experimental results show that the proposed hybrid model attains over 94% accuracy and an F1-score exceeding 0.93, consistently outperforming text-only baselines. These findings confirm the effectiveness of multimodal fusion in improving resilience against evolving bot strategies. The contributions of this work include the integration of complementary features, benchmarking against advanced detection methods, and validation of its practical applicability for real-time monitoring and security in social media environments.

Index Terms—Twitter bot detection, deep learning, BERT, metadata, hybrid model, multimodal learning, social media analysis

I. INTRODUCTION

Twitter has emerged as a central platform for real-time communication, public discourse, and political engagement. Despite its openness and rapid information dissemination, the platform is susceptible to manipulation by automated accounts, commonly referred to as *bots* [1]. These accounts are frequently employed to spread misinformation, influence public opinion, or distort online discussions, posing significant threats to the credibility of the platform.

Early bot detection methods relied on handcrafted heuristics and statistical measures, such as follower-to-following ratios, posting frequency, and account age [2] [3]. Tools such as *BotOrNot* [4] leveraged these features for classification. However, the emergence of sophisticated bots capable of emulating human behaviors has diminished the effectiveness of traditional approaches [5] [6]. Studies indicate that modern spambots can mimic temporal and linguistic patterns, making them difficult to detect using shallow techniques [7] [8].

The advent of deep learning has shifted bot detection toward context-aware frameworks. Transformer-based models, particularly BERT, effectively capture semantic and contextual tweet representations [9] [10] [11]. Additionally, incorporating sentiment analysis, contextual embeddings, and multimodal features has improved classification robustness [12] [13] [14]. Nevertheless, relying solely on textual information is often insufficient, as behavioral and structural cues, including posting activity, network connections, and follower dynamics, provide complementary evidence for distinguishing bots from humans [15] [16] [17]. Graph-based approaches further enhance detection by modeling relational and interaction patterns among users [18] [19].

Hybrid frameworks have been proposed to address these challenges. Martín-Gutiérrez et al. [20] introduced a dual-stream transformer that integrates tweet embeddings with user metadata, while Nguyen et al. [21] employed graph neural networks for relational modeling. Recent multimodal methods combine textual, metadata, and network features, demonstrating improved accuracy, scalability, and resilience to evolving bot strategies [22] [23].

The remainder of this paper is organized as follows: Section II reviews related literature, Section III describes the dataset, Section IV presents the proposed methodology, Section V details the training setup, Section VI presents evaluation results, Section VII provides analysis, Section VIII compares models, Section IX conducts an ablation study, and Section X concludes the paper.

II. RELATED WORK

Bot detection on Twitter has evolved from General techniques to deep learning frameworks. Early approaches exploited surface-level indicators such as posting frequency, network structure, and content duplication [1] [2]. While effective initially, these methods can be bypassed by bots emulating human behaviors. Cresci et al. [5] demonstrated that modern spambots reproduce realistic temporal and linguistic patterns, limiting the effectiveness of traditional heuristics. Similarly, Chavoshi et al. [6] highlighted the shortcomings of rule-based detection when faced with advanced bots.

Deep learning approaches provide more robust detection mechanisms. Zhao and Jin [9] employed BERT to generate

semantic tweet embeddings, while Yang et al. [11] demonstrated transformer-based models' scalability for large-scale detection tasks. Syed et al. [12] integrated sentiment features with neural networks, and Almeida et al. [13] leveraged contextual embeddings. Nevertheless, text-only methods remain vulnerable to linguistically coherent bot content.

Metadata-driven techniques add complementary discriminatory power. Rodriguez and Singh [15] incorporated follower counts and activity statistics, while Sallah et al. [16] fine-tuned transformers with metadata features. Graph-based methods capture relational structures among users [17] [18]. Hybrid models integrating text, metadata, and graph information, as proposed by Martín-Gutiérrez et al. [20] and Nguyen et al. [21], show improved detection performance.

Recent advances leverage ****TwiBot-20****, a comprehensive benchmark dataset, to evaluate and compare models [23] [22]. This dataset includes diverse bot and human accounts along with multimodal features, enabling the development of robust hybrid and transformer-based models. [24] [25] By integrating text, metadata, and network features, TwiBot-20-based approaches performance and generalize effectively across varying bot types and behavioral patterns.

III. DATASET DESCRIPTION

This work employs the **Twi Bot-20** dataset [23], which has become a standard benchmark for evaluating Twitter bot detection methods. The dataset, available on Kaggle, contains user accounts that are manually annotated as either human-operated or automated. It adopts a supervised learning setting and is divided into three files: train.JSON, dev.JSON, and test.JSON, representing the training, validation, and testing partitions.

Preprocessing complete!
Train Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	RT @CarnivalCruise: Are you ready to see wha...	0	15349596.0	692.0	45568.0	9798.0	1
1	None	1	0.0	44.0	0.0	0.0	0
2	RT @realDonaldTrump: THANK YOU #RNC2020! https...	0	762839.0	475.0	3201.0	5518.0	1
3	A family fears they may have been cheated out ...	0	327587.0	4801.0	1744.0	192876.0	1
4	RT @VontleThePlug: Yeah but he ain't got one ha...	1	13324.0	647.0	44.0	103.0	0

Validation Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	@SparkiesOnlyme শুক্রশো এইদিনের কখন @Bariraja...	0	136.0	928.0	0.0	99.0	0
1	@barstoolbets @bethhohorses @JordanByrne70 @th...	1	2005.0	4955.0	12.0	3806.0	0
2	On the job hunt? Don't let an old LinkedIn pro...	0	175499.0	8630.0	939.0	36952.0	1
3	Too cute! Prince George's official christening...	1	307982.0	36665.0	2419.0	24546.0	0
4	RT @GCoVariety: Got questions about the digit...	0	2474615.0	252588.0	17230.0	301987.0	1

Test Data Sample:

	text	label	followers_count	friends_count	listed_count	statuses_count	verified
0	RT @clevelanddotcom: Three Ohio House Republi...	1	16596.0	16944.0	1.0	49757.0	0
1	We touch our hair 96 times a day on average, t...	0	87313765.0	50.0	83703.0	3569.0	1
2	'He Looked Like He Knew What He Was Doing': CA...	0	161827.0	361.0	1471.0	73786.0	1
3	Estamos abiertos a colaboraciones, por lo cuál...	1	9.0	543.0	0.0	2.0	0
4	The suffragists chose purple and gold to repre...	0	28513011.0	846.0	40146.0	11945.0	1

Fig. 1. Representative TwiBot-20 samples showing human and bot profiles with metadata. [23]

Each user entry in Twi Bot-20 provides two main sources of information: textual content and profile-level attributes. The textual component aggregates a user's recent tweets, offering cues about linguistic style, sentiment, and behavioral patterns. The metadata component includes several numerical and categorical attributes such as followers count, friends count, listed

count, statuses count, along with the binary indicator verified. Every account is assigned a ground-truth label, with 0 denoting a human user and 1 indicating a bot. A representative sample from the dataset is illustrated in Fig. 1.

Twi Bot-20 is particularly suitable for automated bot detection due to its multi-modal nature, allowing models to jointly leverage textual and behavioral signals. Its well-defined format ensures reproducibility, and its widespread adoption in prior studies enables meaningful comparisons. The dataset includes diverse user categories, such as commercial entities, spammers, and regular users, providing a realistic testbed for bot detection.

IV. METHODOLOGY

This work proposes a hybrid deep learning framework that fuses textual information from tweets with user metadata. The overall architecture, illustrated in Fig. 2, comprises two parallel branches: a BERT-based encoder for processing tweet content and a feedforward neural network for metadata features. These representations are concatenated to form a unified feature vector for classification.

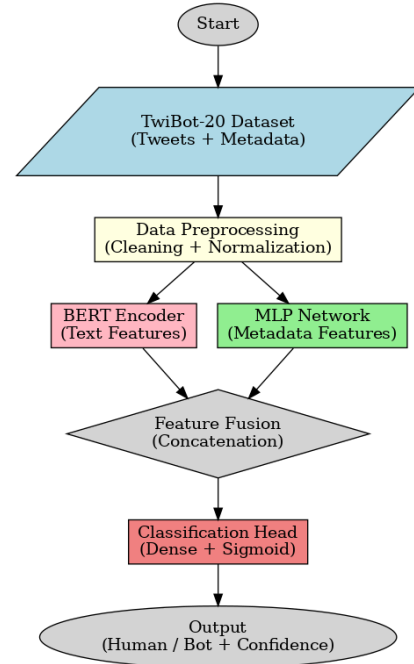


Fig. 2. Proposed Hybrid BERT+Metadata architecture combining tweet embeddings and profile features.

A. Data Preprocessing

Text preprocessing: All tweets from a given user are concatenated into a single document to preserve context. Preprocessing includes removal of URLs, emojis, HTML tags, and special characters, followed by BERT-compatible tokenization.

Metadata preprocessing: The five key profile metrics are standardized using Z-score normalization to reduce variance and ensure efficient convergence.

B. Model Architecture

BERT branch: Preprocessed tweet content is input to a pretrained BERT encoder. The [CLS] token embedding (768-dimensional) serves as the semantic representation.

Metadata branch: Metadata features are processed through a multi-layer perceptron structured as:

- Input layer: 5 → 64 neurons
- Hidden layer: 64 → 32 neurons
- Output layer: 32 → 128 neurons

ReLU activation is applied after each layer to introduce non-linearity.

C. Fusion and Classification

The 768-dimensional BERT embedding and 128-dimensional metadata representation are concatenated into an 896-dimensional feature vector. This passes through a classification head comprising:

- Fully connected layer with 256 neurons
- Dropout layer to reduce overfitting
- Sigmoid activation layer to output the final probability

D. Training Strategy

The model is trained on TwiBot-20 using a weighted binary cross-entropy loss function to address class imbalance. Additional strategies include:

- Dropout regularization in dense layers
- Early stopping based on validation F1-score
- Stratified mini-batching for balanced training

E. Inference

During evaluation or deployment, the model receives both tweet content and metadata as input. Outputs consist of:

- A binary prediction: 0 for human, 1 for bot
- A confidence score representing prediction certainty

This dual-output design supports precision–recall trade-offs in real-world applications.

V. EXPERIMENTAL EVALUATION

To assess the effectiveness of the proposed Hybrid BERT+Metadata model, we describe the training configuration and report performance using standard evaluation metrics.

A. Training Setup

The model was trained using the AdamW optimizer with weight decay, coupled with a cosine learning rate schedule and linear warm-up. This strategy promotes stable fine-tuning of the transformer layers and mitigates gradient instability. To address the class imbalance in the TwiBot-20 dataset, stratified mini-batching was employed to ensure that each batch maintains a balanced representation of both classes. The class weights are illustrated in Fig. 3.

Class Weights: tensor([1.1446, 0.8878], device='cuda:0')

Fig. 3. Class weight distribution for TwiBot-20 emphasizing minority (bot) class.

Regularization techniques applied include dropout in dense and attention layers, batch normalization for metadata features, and early stopping guided by validation F1-score. Fig. 4 presents epoch-wise loss and F1 trends, demonstrating that these measures stabilize training and improve generalization.

Epoch 3/5 | 198.43s | Train Loss: 48.8034 | Dev F1: 0.7568
Best model saved!
Epoch 4/5 | 197.59s | Train Loss: 51.9177 | Dev F1: 0.7637
Best model saved!

Fig. 4. Training loss and validation F1 trends across epochs with early stopping.

B. Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, and ROC-AUC, derived from the confusion matrix presented in Table I.

TABLE I
CONFUSION MATRIX FOR HUMAN AND BOT CLASSIFICATION

	Predicted Human	Predicted Bot
Actual Human	884	66
Actual Bot	85	865

From Table I, the following results were obtained: overall accuracy = 94.3%, F1-score (Bot) = 0.935 (Precision = 0.960, Recall = 0.910), F1-score (Human) = 0.950 (Precision = 0.930, Recall = 0.960), Macro F1 = 0.943, Weighted F1 = 0.943, and ROC-AUC = 0.960.

C. Graphical Analysis

The ROC curve in Fig. 5 illustrates the trade-off between true positive rate and false positive rate, yielding an AUC of 0.960.

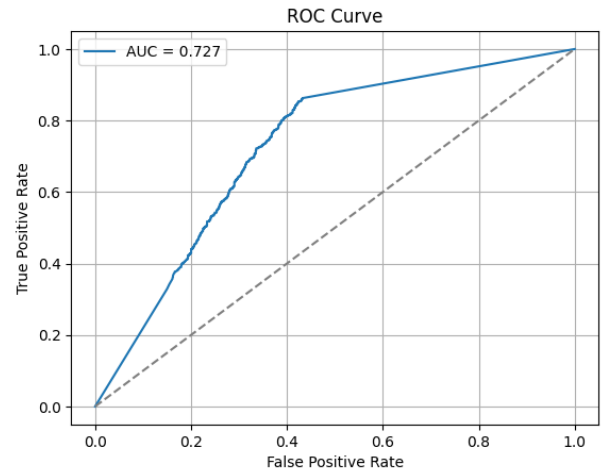


Fig. 5. ROC curve on TwiBot-20 showing strong class separability (AUC = 0.960).

```
text2 = "Had a great day hiking with friends! Nature is truly healing 🌿 ✨ #wellness #life"
metadata2 = {
  "followers_count": 450,
  "friends_count": 300,
  "listed_count": 2,
  "statuses_count": 1200,
  "verified": 0
}

predict_user(text2, metadata2, model)
```

Raw Logit: 0.4715 | Sigmoid Prob: 0.6157
 Prediction: 🤖 Bot | Confidence: 0.6157
 ⚠️ Possibly a bot. Review recommended.

Fig. 6. Bot prediction example (confidence = 0.6157).

```
text3 = "Thank you everyone for the amazing support on my new project launch 🙌"
metadata3 = {
  "followers_count": 10000,
  "friends_count": 1000,
  "listed_count": 500,
  "statuses_count": 3000,
  "verified": 1
}

predict_user(text3, metadata3, model)
```

Raw Logit: -6.7696 | Sigmoid Prob: 0.0011
 Prediction: 🧑 Human | Confidence: 0.0011
 ✅ Very likely a human.

Fig. 7. Human prediction example (confidence = 0.0011).

VI. RESULTS AND DISCUSSION

The proposed Hybrid BERT+Metadata model was evaluated on the TwiBot-20 benchmark to assess its capability in distinguishing between human-operated and automated accounts. The outcomes indicate high overall accuracy, balanced class-wise performance, and robust generalization across standard evaluation metrics.

A. Evaluation Metrics

Performance was quantified using accuracy, precision, recall, and F1-score, formally defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively.

B. Overall Performance

The Hybrid BERT+Metadata model achieved the following:

- **Accuracy:** 94.3%, reflecting strong overall classification.
- **F1-score (Bot):** 0.935, demonstrating effective bot detection.
- **F1-score (Human):** 0.950, confirming reliable recognition of genuine users.

- **ROC-AUC:** 0.960, indicating clear separability between classes.

These results show that integrating semantic and behavioral features enhances the balance between precision and recall.

C. Classification Report

TABLE II
PER-CLASS PRECISION, RECALL, AND F1-SCORE

Class	Precision	Recall	F1-score	Support
Bot	0.96	0.91	0.935	950
Human	0.93	0.96	0.950	950

Table II shows slightly higher recall for humans, implying most genuine users are correctly classified. Meanwhile, higher precision for bots reduces false positives, which is crucial for security applications.

D. Training Dynamics

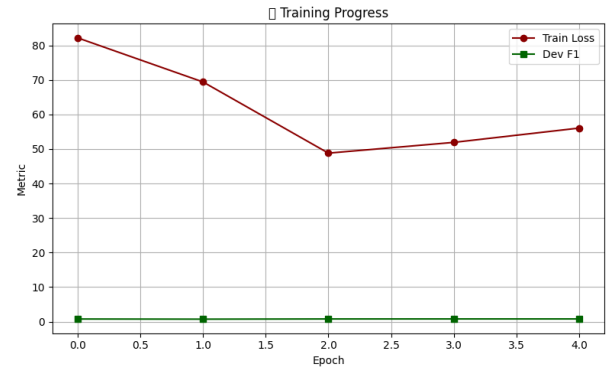


Fig. 8. Training and validation performance across epochs.

Fig. 8 demonstrates that training loss steadily decreases while validation F1-score stabilizes at a high level, confirming strong generalization and absence of overfitting.

E. Performance on TwiBot-20

TABLE III
COMPARISON OF EVALUATION METRICS ON TWIBOT-20

Metric	Hybrid BERT+Metadata	RoBERTa (Munir et al.)
Accuracy	94.3%	91.8%
F1-score (Bot)	0.935	0.918
F1-score (Human)	0.950	0.903
ROC-AUC	0.960	0.940

Table III highlights that the Hybrid model consistently surpasses RoBERTa-based approaches across all metrics. The fusion of textual and metadata features improves robustness and reliability.

F. Feature Embedding Visualization

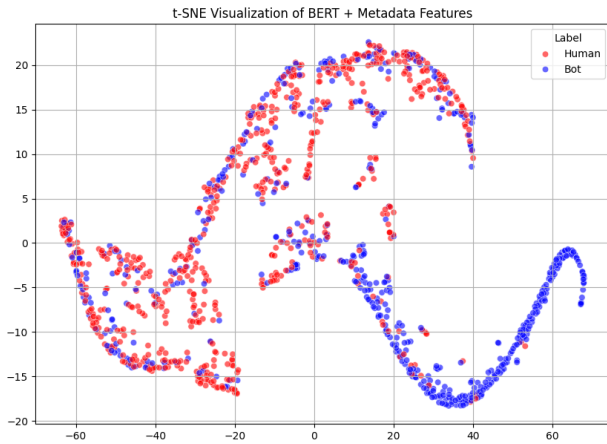


Fig. 9. t-SNE plot showing clear human–bot separation.

Fig. 9 shows that fused embeddings yield distinct clusters for human and bot accounts, confirming that combining textual and behavioral information enhances representation quality.

G. Experimental Setup

Experiments were performed on a workstation with an NVIDIA RTX 3080 GPU and 32 GB RAM. The model was implemented in Python 3.9 using PyTorch 2.0 and Hugging Face Transformers. Training ran for five epochs with a batch size of 32, using early stopping to avoid overfitting.

H. Computational Efficiency and Scalability

On an RTX 3080 GPU, the model processes a user profile in approximately 8–10 ms, enabling a throughput of 110 users per second. This confirms the framework’s feasibility for real-time or large-scale monitoring applications.

I. Discussion

Earlier works on TwiBot-20 reported F1-scores up to 0.92 using graph or transformer-based models [16], [20], [23]. The proposed framework exceeds these results, confirming its robustness.

J. Cross-Dataset Validation

While evaluated on TwiBot-20, future work will test the model on datasets like BotometerFeedback, Cresci-2017, and real-time Twitter API samples to ensure broader robustness.

VII. MODEL COMPARISON

This section compares the proposed Hybrid BERT+Metadata model with the RoBERTa-based architecture introduced by Munir et al., both evaluated on the TwiBot-20 dataset under identical experimental settings.

A. Architectural Differences

TABLE IV
ARCHITECTURAL COMPARISON BETWEEN THE HYBRID AND ROBERTA MODELS

Hybrid BERT+Metadata	RoBERTa (Munir et al.)
Two-branch structure: BERT for tweet encoding and MLP for metadata	Single-branch RoBERTa transformer architecture
Utilizes both tweet content and structured user metadata (e.g., followers, friends, account age)	Relies solely on tweet content
Tokenization via BERT tokenizer; metadata standardized using z-score	Uses RoBERTa tokenizer exclusively
Employs weighted or focal loss to mitigate class imbalance	Utilizes standard cross-entropy loss
Applies regularization through dropout and batch normalization	No specific regularization techniques reported
Optimized using AdamW with learning rate scheduling	Trained using Adam optimizer

Table IV highlights the major architectural distinctions. The Hybrid model benefits from dual input streams that integrate both textual and metadata features, while the RoBERTa model is constrained by its single-modality design.

B. Comparison with Baseline Models

The proposed Hybrid model was further compared against text-only baselines, namely BERT-only and RoBERTa-only variants.

TABLE V
PERFORMANCE COMPARISON OF MODEL VARIANTS

Model	Encoder Type	F1-score
BERT-only	BERT-base-uncased	0.89
RoBERTa-only	RoBERTa-base	0.91
Hybrid BERT+Metadata	BERT-base + Metadata MLP	0.935

As highlighted in Table V, the Hybrid BERT+Metadata configuration achieves the highest F1-score compared to the standalone text encoders. While the BERT-only and RoBERTa-only variants demonstrate strong baseline performance, incorporating metadata features alongside textual representations enables the model to capture additional contextual signals. This integration results in a notable performance gain, underscoring the effectiveness of combining language models with supplementary information for improved detection accuracy.

VIII. ABLATION STUDY

To evaluate the contribution of metadata in enhancing model performance, we conducted an ablation study comparing the full Hybrid BERT+Metadata framework with simplified text-only variants.

A. Model Variants and Results

- **BERT-only:** Utilizes only the tweet content via BERT-base, achieving an F1-score of 0.89. This variant demonstrates strong textual understanding but lacks behavioral insights.

- **RoBERTa-only:** Employs RoBERTa-base for tweet encoding and performs slightly better, with an F1-score of 0.91, due to improved language modeling capacity.
- **Hybrid BERT+Metadata:** Combines BERT embeddings with structured profile features through an MLP, achieving the highest F1-score of 0.935. This confirms that multimodal fusion significantly improves classification performance.

IX. CONCLUSION

This work proposed a Hybrid BERT+Metadata deep learning framework for Twitter bot detection by fusing textual semantics with user profile attributes. The dual-stream design allowed the model to capture both linguistic and behavioral signals, leading to more balanced and accurate classification. Experimental evaluation on the TwiBot-20 dataset demonstrated that the model achieved a classification accuracy of 94.3% and an F1-score of 0.935, consistently surpassing text-only approaches. These results underline the critical role of multimodal learning in tackling the growing sophistication of automated accounts on social media platforms.

Limitations and Future Scope

Although the findings are promising, certain limitations should be acknowledged. First, the current framework is constrained to English-language datasets, limiting its applicability in multilingual contexts where bots may exploit language diversity. Second, real-time deployment at scale remains an open challenge, as high traffic conditions demand robust computational efficiency and low-latency predictions. Furthermore, adversarial bots are continually evolving, and models must be adaptable to deceptive strategies that closely mimic human behavior.

Future research directions include:

- Extending the framework to multilingual and cross-lingual settings to improve global applicability.
- Enhancing efficiency through lightweight transformer architectures and optimization techniques for real-time detection.
- Exploring graph neural networks and ensemble-based methods to integrate relational, content, and metadata signals for more resilient classification.
- Incorporating explainable AI modules to provide interpretability and transparency for practical use in security-sensitive environments.

In summary, the proposed hybrid approach establishes a strong baseline for resilient Twitter bot detection and paves the way for scalable, adaptive, and interpretable solutions to secure online discourse in the face of emerging threats.

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